

UTILIZING MAXENT TO DETERMINE THE TOP COVARIATES RELATED TO MYOTIS ACTIVITY IN CENTRAL WISCONSIN

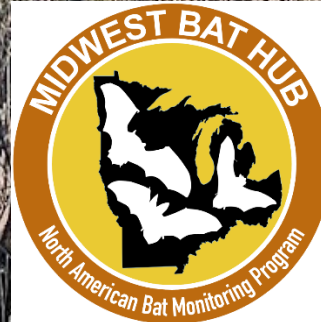
PRESENTED BY SAM COLEMAN

CO-AUTHORS: GAGE KRIESE, GABBY SHULFER,
ELIZABETH GROVE

ADVISOR: DR. CHRIS YAHNKE



Wisconsin Center for Wildlife
College of Natural Resources
University of Wisconsin - Stevens Point



MYOTIS IN WISCONSIN

- Little brown bat (*Myotis lucifugus*) and the Northern long-eared bat (*M. septentrionalis*)
- Both are state threatened, N. long-eared is federally listed as Endangered
- Knowledge gaps surrounding habitat selection in *Myotis*



Credit: Northwest Wildlife Preservation Society



Credit: Newport Wilderness Society

MYOTIS LIFE HISTORY

- Roost in trees, caves, attics and barns
- Insectivores, pregnant females can eat up their body weight in mosquitos in one night
- Females live in maternity colonies during the summer
 - Roost in cavities or under peeling bark of deciduous trees and snags during the summer
 - Pups begin flight around July
- Tree roost selection based on:
 - Presence of shaggy bark
 - DBH
 - Decay stage
 - Canopy class
 - Species



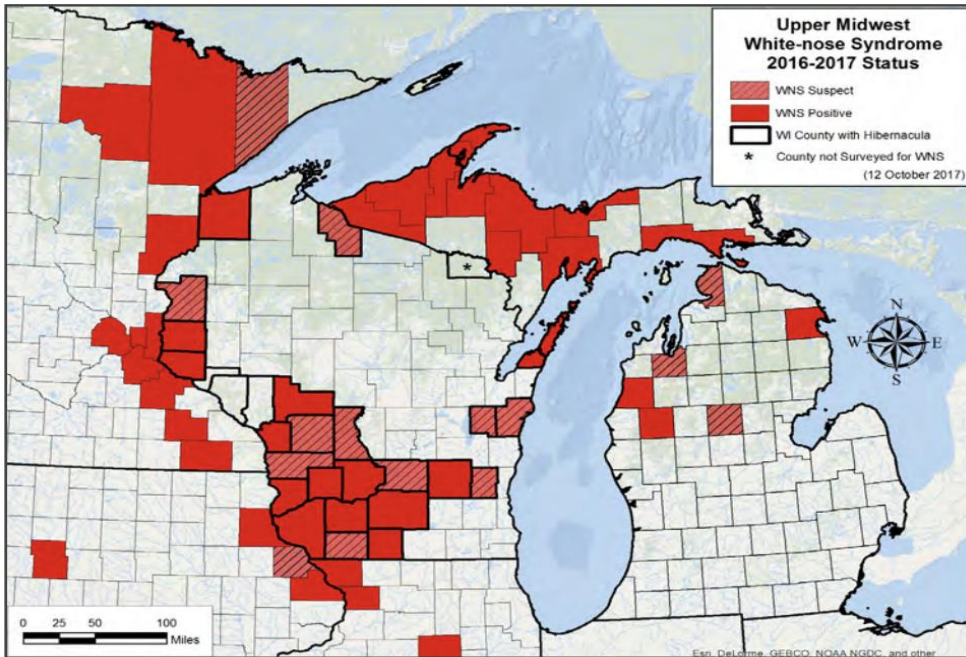
Credit: Four Seasons Tree Care



Credit: Carl Morrow

MYOTIS AND WHITE NOSE SYNDROME

- White-nose syndrome is a concern for hibernating bats
 - Caused by the fungus *Pseudogymnoascus destructans*
 - Reduce cave bat colonies by up to 95%
 - First documented to Grant County, Wisconsin in 2014



Credit: Wisconsin Bat Program, WI DNR



Credit: Marvin Moriarty, USFWS

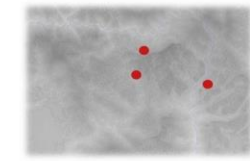
SPECIES DISTRIBUTION MODELING

- Species distribution modeling helps identify potential areas for species occurrences
- Utilizes environmental data to predict spatial distribution of species
- Can be used to investigate what habitats species are selecting for
- Many studies use SDM on a large species-range wide scale
- Few studies on local areas

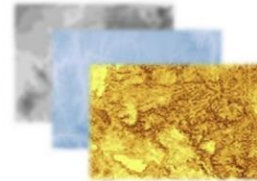
Searching for a rare species...



...correlating...

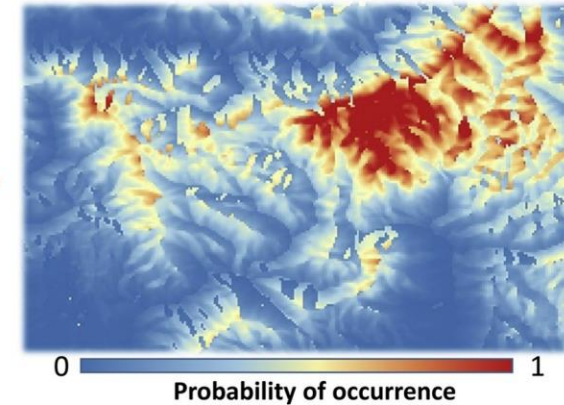


Limited information on species occurrence

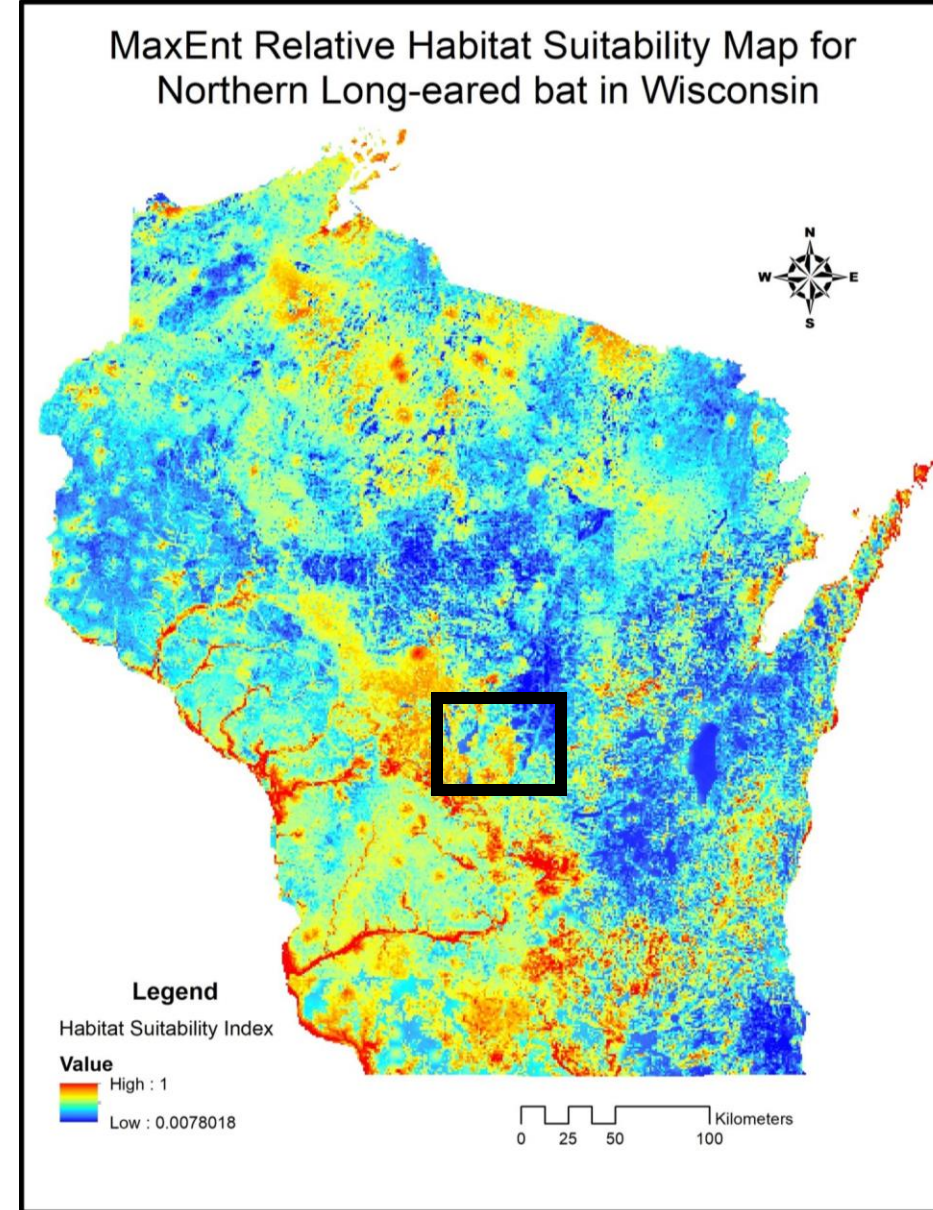


Environmental factors

Using distribution models' outputs to guide field surveys



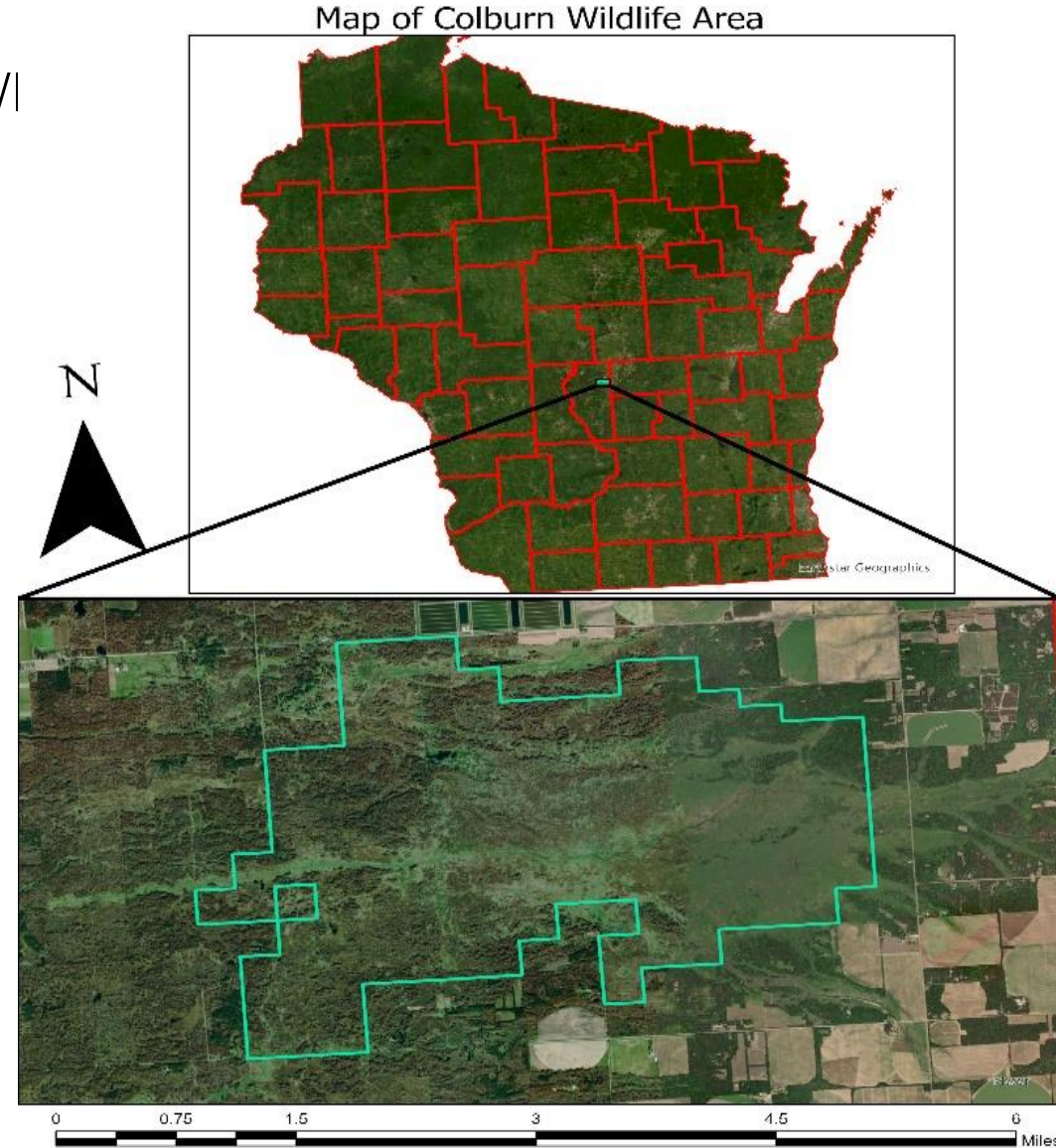
Credit: Fois et al. 2018



Kaarakka, H. 2019. Summer Distribution and Habitat Suitability for Two Threatened Bat Species in Wisconsin. [Master's Thesis, University of Wisconsin – Madison]

COLBURN WILDLIFE AREA

- 4,965 acres (2009 hectares), located in Adams County, WI
- Avg 34 inches of rain per year
- High temps around 74°F-82°F (Summer 2024)
- Low temps around 52°F-61°F (Summer 2024)
- Important wildlife area providing habitat for ruffed grouse, woodcock, waterfowl, songbirds, white-tailed deer, and bats



ASPEN WETLANDS

- 47% of the property
- Reed canary grass, sensitive fern, cattails, quaking aspen, willows, greater water dock and gray alders
- Interspersed oak upland islands throughout large swaths of connected reed marshes
- Average of 31 trees per acre



OAK UPLANDS

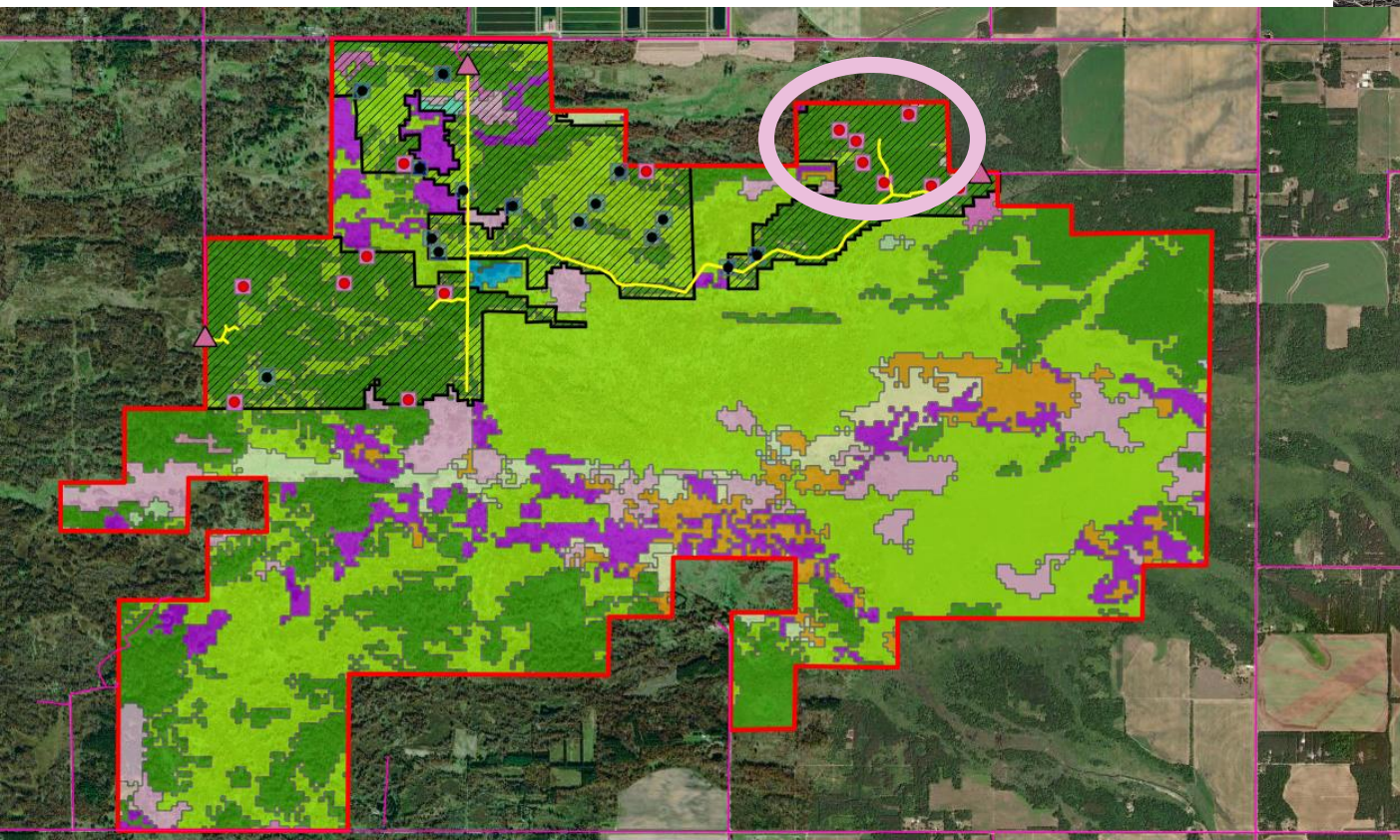
- Takes up 32% of the property
- Dominated by black oak and N. pin oak, red and silver maples, Pennsylvania Sedge, bracken fern, sweet-ferns, and wood ferns
- Interspersed wildflowers including wild bergamot, Michigan lilies, butterfly milkweed, Carolina roses and bird's foot violet
- Average of 60 trees per acre



UNTOUCHED MANAGEMENT APPROACH

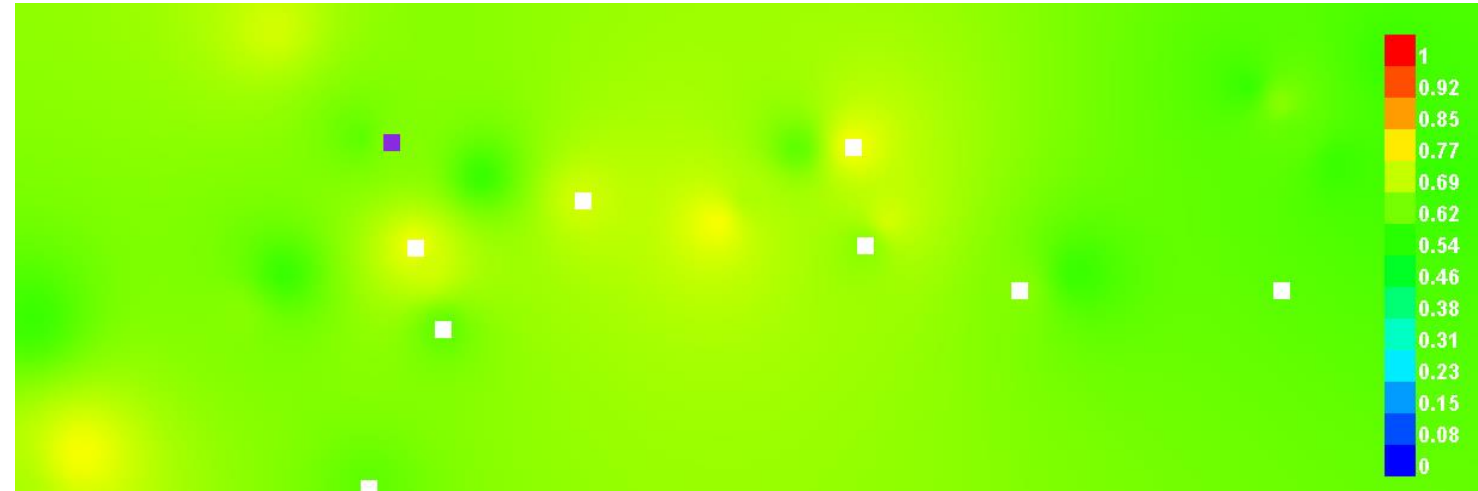


LOGGING PROJECT IN NORTHWEST CORNER



MAXENT

- Maximum Entropy theory
- Accounts for nature's stochasticity
- Presence-only data
- Critical for species conservation
- Predicts the highest suitable habitat ranges
- Uses "training" points from occurrence data to test model
- Uses "background" points across entire study area to simulate absences



Purple point is training point
Background points identify suitable habitat

WHAT WAS MEASURED IN THE FIELD?

- Variable radius plots were conducted in randomly generated points with BAF 10 prisms
- Recorded each "in" tree
- Every other borderline tree was measured
- 15 in Aspen wetlands - 15 in Oak stands
- Each tree was measured for:
 - DBH
 - Canopy Class
 - Decay Stage (1-5)
 - Presence or absence of peeling bark
 - Tree species

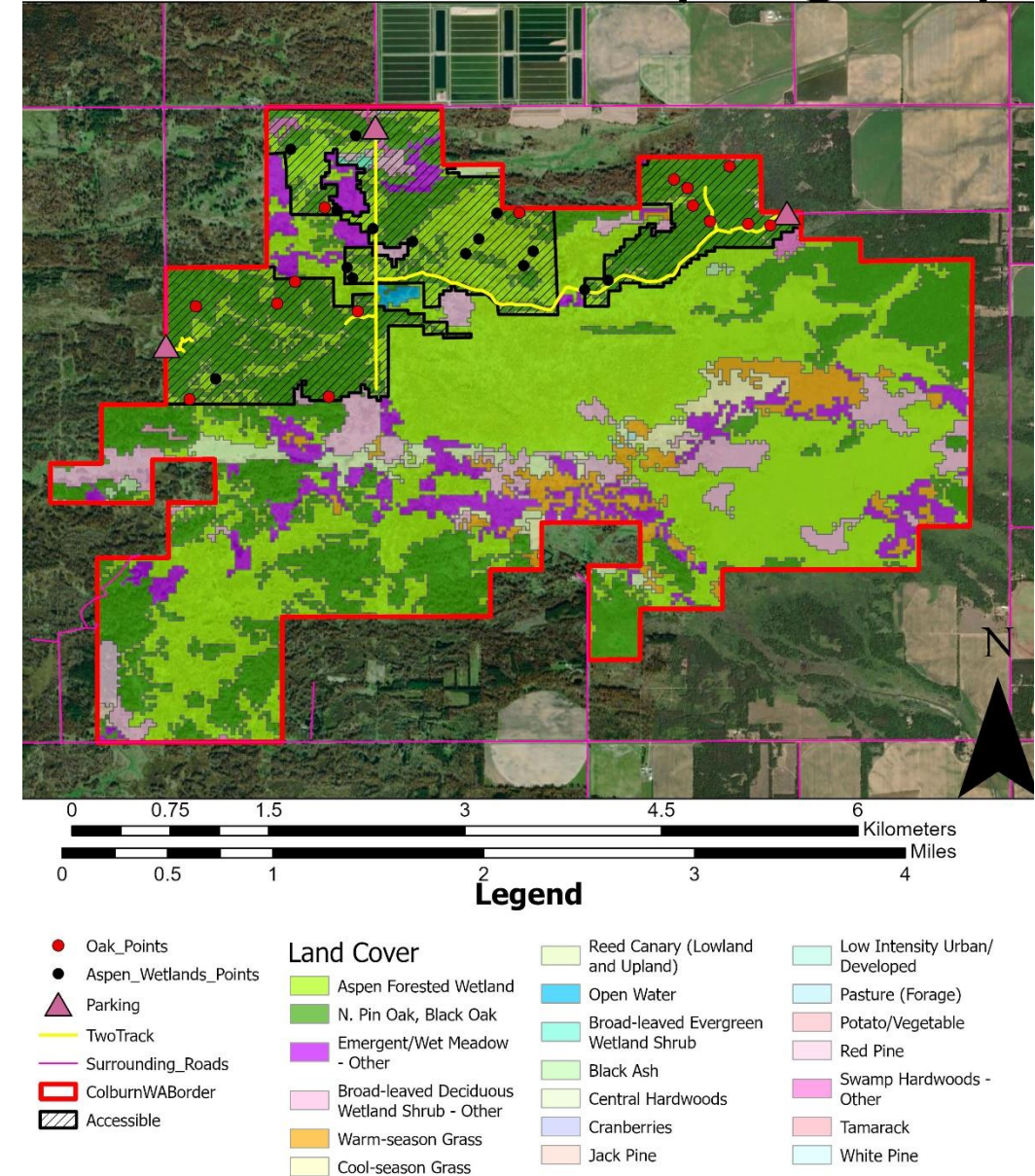


Credit: Stephen Pohlman

SAMPLING POINTS

- "Accessible" areas were determined by scouting out Colburn
- Deemed for safety of volunteers and to keep within scope of our project
- Used Wisclands 2.0 level 4 data (60% accurate)
- ArcGIS pro used to randomly select 15 points in aspen wetlands and 15 oak stands within accessible area
- 100m buffer zone around each point

Colburn Tree Sampling Map



FIELD MAPS



- Navigate to VRPs – compass feature
- Field volunteers entered tree data on personal devices
- Points get uploaded to an online spreadsheet once connected to service
- Makes it easy to see who measured which tree

COVARIATES/ENVIRONMENTAL LAYERS FOR MAXENT

- Each plot's basal area
- Each plot's basal area of trees with shaggy bark
- Each plot's average of tree's decay stage (scale of 1-5)
- Each plot's average of ordinaly rated canopy class
- Each plot's rated score of aspen-oak ratio
 - 80-100% aspen = 1
 - 50-80% aspen = 2
 - Neither aspen nor oak were dominant = 3
 - 50-80% oak = 4
 - 80-100% oak = 5



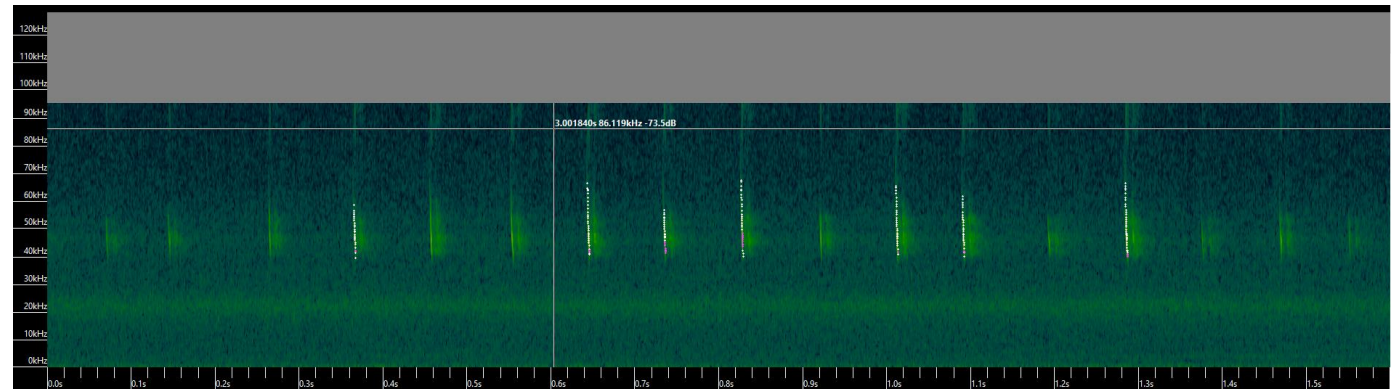
BAT DETECTORS

- Deployed four AudioMoth portable bat detectors in randomly selected VRP points every 14 days
- Detectors were filtered to only detect *Myotis spp.* calls to save space on microSD cards
- Without filtering, detectors would only record for 4 days, with filtering 10 days

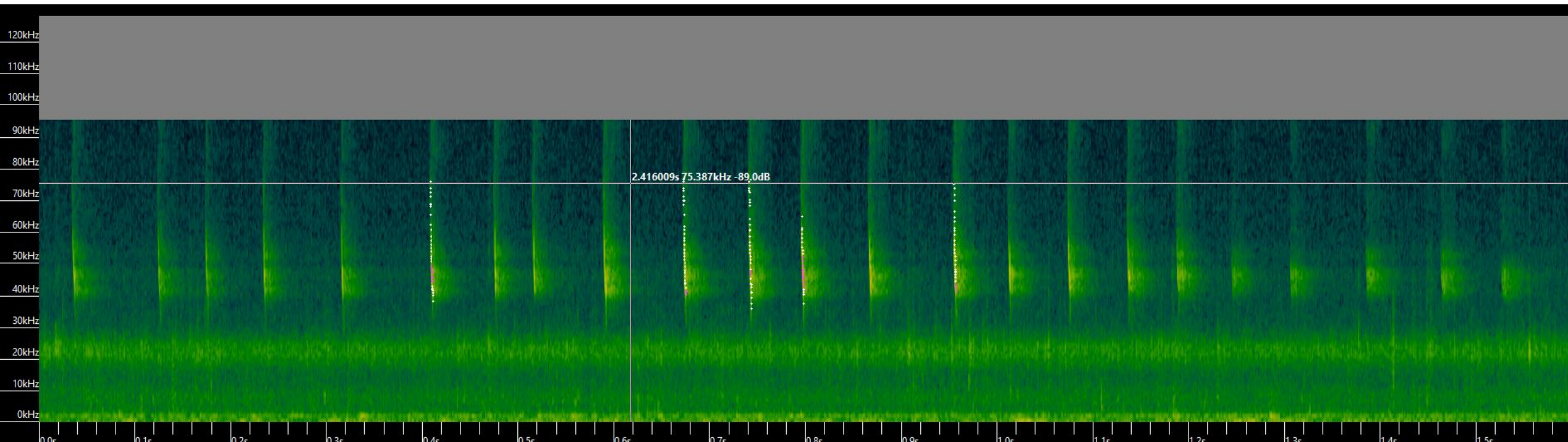


ANALYZING BAT CALLS

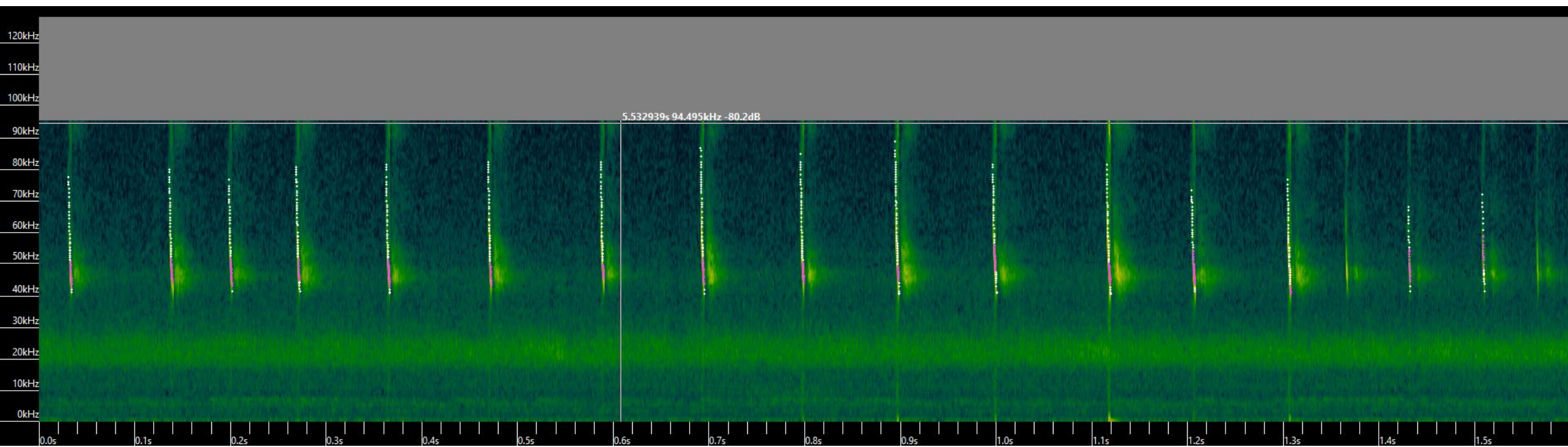
- WAV files were uploaded to a shared Google Drive
- Batches of files were analyzed with Kaleidoscope Pro
- Trace back the detector name and date the bat call was detected to detector deployment location



Northern long-eared bat (*Myotis septentrionalis*) sonogram. Frequency ranges from 40 kHz-68 kHz



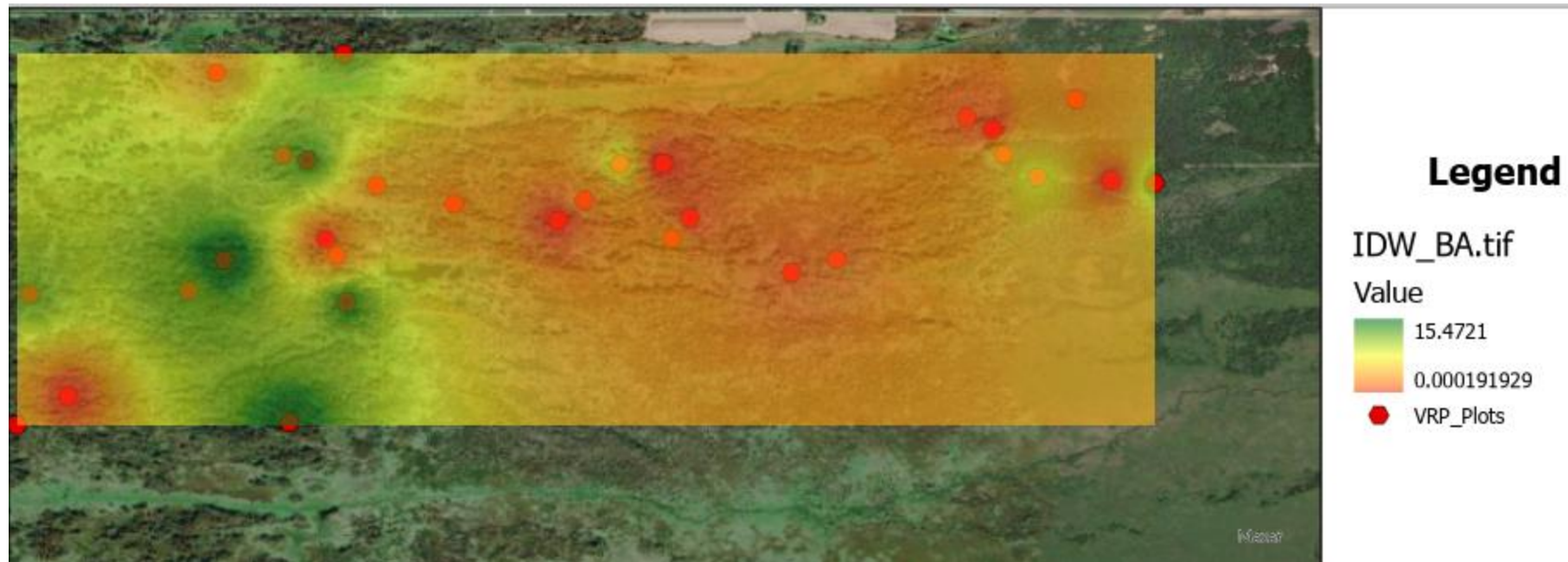
Northern long-eared bat (*Myotis septentrionalis*) sonogram. Frequency ranges from 35 kHz-75 kHz



Northern long-eared bat (*Myotis septentrionalis*) sonogram. Frequency ranges from 40 kHz-89 kHz

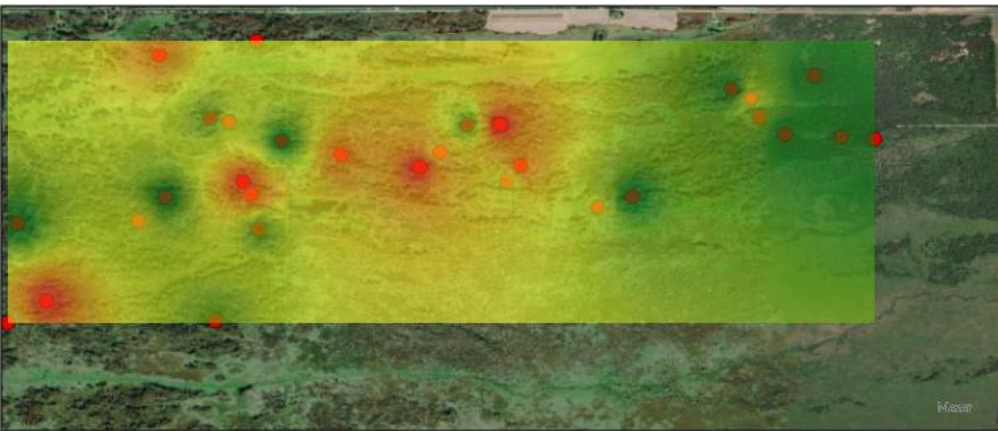
RASTERIZING THE DATA

- ArcGIS Pro to convert each VRP to a vector point
- QGIS Inverse distance weighted (IDW) interpolation
- Each cell has a value based on information from the nearest VRP
- Raster TIFs were converted to ASCII files for MAXENT

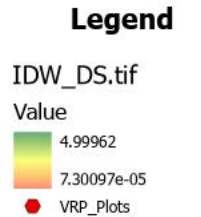
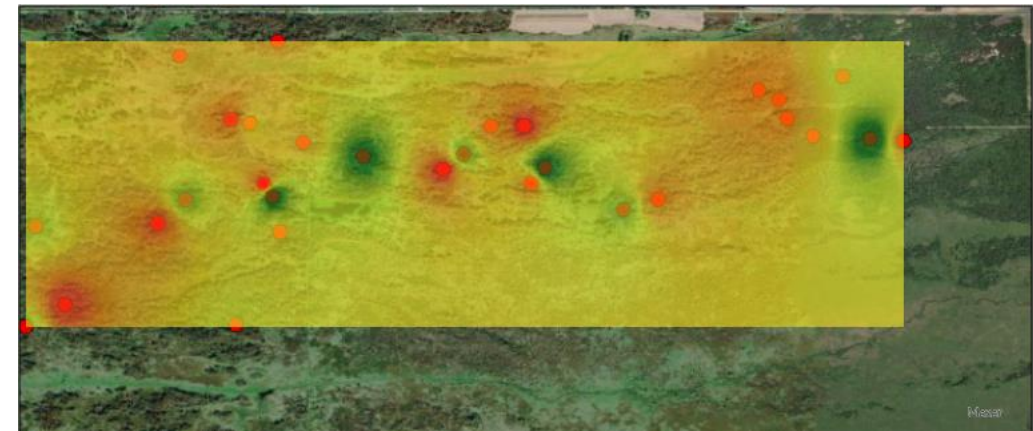


Inverse distance weighted interpolation map of each plot's basal area

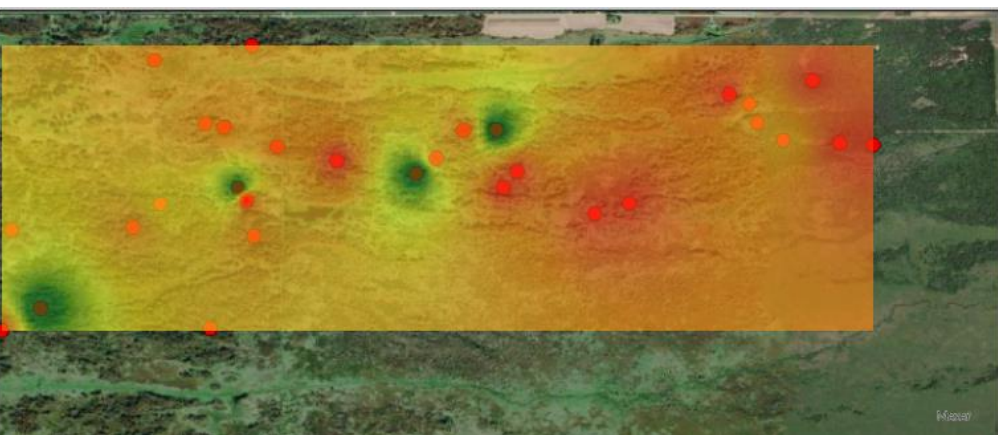
RASTERIZED MAPS



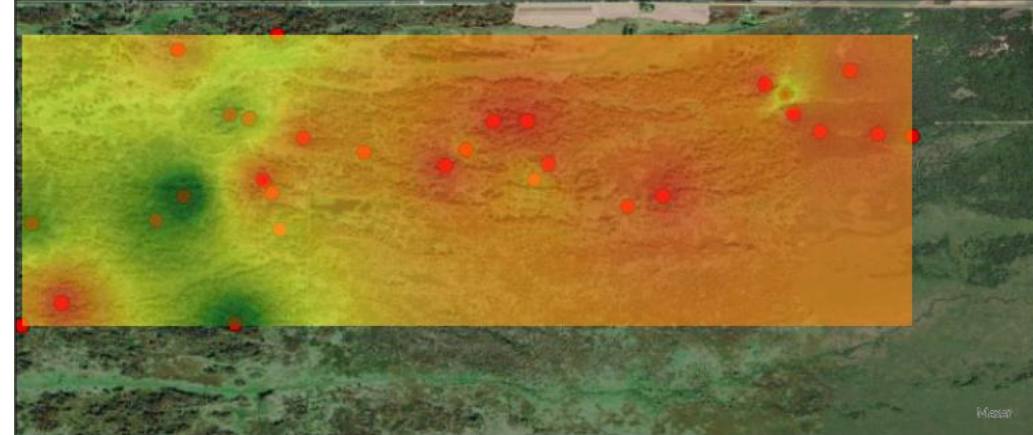
Aspen-Oak Scale (1-5)



Average Plot Decay Stage (1-5)

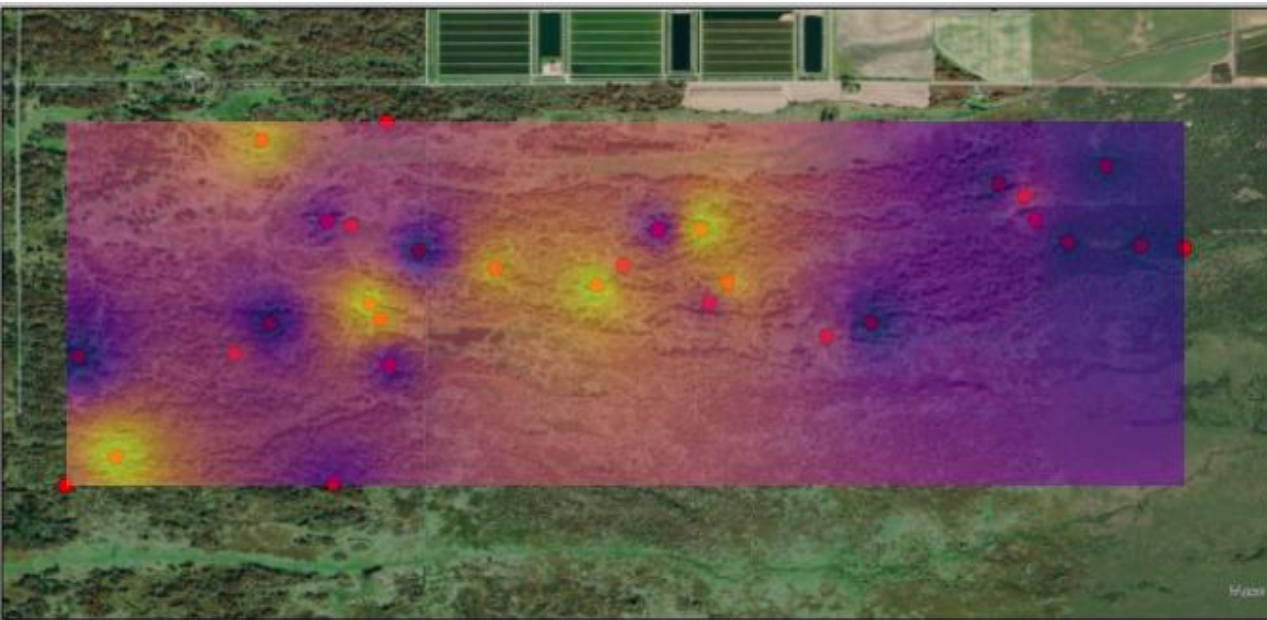


Average Plot Canopy
Class (1-4)



Plot's BA of trees with
Shaggy Bark

HEATMAPS OF MYLU AND MYSE



Legend

MYLU.asc

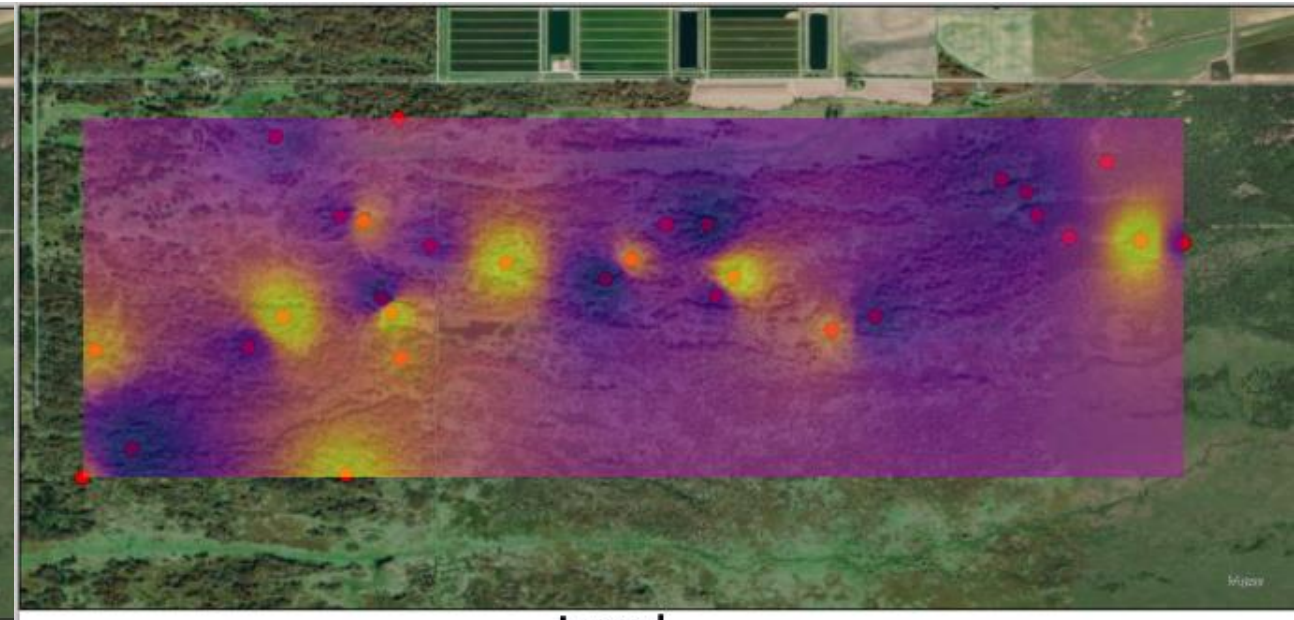
Value

0.741606

0.555706

VRP_Plots

Little Brown Bat (*Myotis lucifugus*)
Habitat Suitability (0 worst – 1 best)



Legend

MYSE.asc

Value

1

0.0439601

VRP_Plots

Northern Long-eared Bat (*Myotis septentrionalis*)
Habitat Suitability (0 worst – 1 best)

MAXENT RESULTS

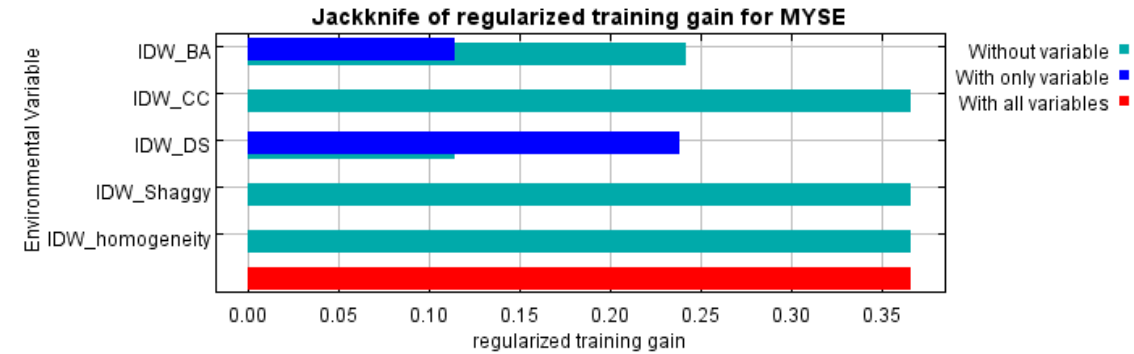
Little Brown Bat (*Myotis lucifugus*)

Variable	Percent contribution	Permutation importance
IDW_homogeneity	100	100
IDW_Shaggy	0	0
IDW_DS	0	0
IDW_CC	0	0
IDW_BA	0	0



Northern Long-eared Bat (*Myotis septentrionalis*)

Variable	Percent contribution	Permutation importance
IDW_DS	66.4	38.2
IDW_BA	33.6	61.8
IDW_homogeneity	0	0
IDW_Shaggy	0	0
IDW_CC	0	0



DISCUSSION

- Only the aspen-oak scale covariate yielded contribution to the MYLU model
- 2 out of 5 covariates, Decay Stage and BA, yielded contribution to MYSE model
- Environmental layers were sub-optimal
- Many detection periods were censored due to battery issues
 - 11/28 successful recording periods
- One detector wasn't filtered properly and resulted in significantly more calls recorded in only 4 days
- Filtering detectors resulted in lower quality recordings and significantly less calls recorded



Map showing all detectors which had successful recording periods

WHAT TO GLEAN FROM THIS DATA

- Tree type could be indicative of where MYLU selects their foraging habitat
- BA and decay stage dictated the MYSE model, but only has 3 occurrence points
- The model was able to identify significant environmental layers tied to both *Myotis* activity



WHAT CAN BE IMPROVED

- Fishnet grid the study area first, then systematically place VRP points
- Record more environmental data (plant community diversity, NDVI/NDWI data, soil type, insect diversity, visual obstruction, distance to woodland edge, etc.)
- Not be reliant on moving detectors from remote areas (if working with volunteers)
- Remove the filtering
- Use IR to delineate biomes instead of WiscLands 2.0
- Integrate more of the study area









ACKNOWLEDGEMENTS

- Wisconsin DNR -Erin Grossman
- UWSP TWS Student Chapter
- Dr. Christopher Yahnke
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- UWSP TWS Bat Project Co-Leaders (Gage Kriese, Gabby Shulfer, Elizabeth Grove, Elize Gabriel and Volunteers
- Wisconsin Center for Wildlife



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QUESTIONS?

